

CIFAR-10 IMAGE CLASSIFICATION

Deep Learning Project Report

CIFARNet & ResNet-50 Transfer Learning

Udacity Machine Learning Nanodegree

PyTorch · Computer Vision · Transfer Learning

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1. Executive Summary

This report presents the design, implementation, and evaluation of a deep learning image classification system trained on the CIFAR-10 benchmark dataset. The work was undertaken as part of the Udacity Machine Learning Nanodegree programme and addresses a real-world decision problem: whether a self-driving car startup should build its own image classification algorithm or purchase one from Detectocorp, an external vendor whose system achieves 70% accuracy on CIFAR-10.

Two models were developed and evaluated. The first, CIFARNet, is a custom-designed compact Residual Network built from scratch using PyTorch. It achieved 93.43% test accuracy on the 10,000-image CIFAR-10 test set after 30 training epochs — exceeding the Detectocorp baseline by 23.43 percentage points. The second model, a ResNet-50 fine-tuned via two-stage transfer learning from ImageNet pre-trained weights, achieved 94.62% accuracy after just 20 epochs of combined training, demonstrating the additional performance benefits of leveraging large-scale pre-trained representations.

Both models substantially outperformed the 70% Detectocorp benchmark and the 45% minimum pass threshold, using state-of-the-art training techniques including data augmentation, batch normalisation, cosine annealing learning rate scheduling, residual skip connections, and differential learning rate fine-tuning. Per-class accuracy analysis revealed consistently strong performance across all ten object categories, with the weakest class (cat, 86.30–88.20%) still far exceeding the competitor baseline.

The clear recommendation arising from this project is to build in-house. The in-house models deliver significantly superior accuracy, provide full architectural control, eliminate recurring licensing costs, and position the organisation to continuously improve performance as proprietary road imagery data is collected from the vehicle fleet. Transfer learning on ResNet-50 further demonstrates that future fine-tuning on domain-specific data can be accomplished efficiently, without retraining from scratch — a critical advantage for a startup operating in a fast-moving autonomous vehicle landscape.

This report details the problem framing, dataset characteristics, model architectures, training methodology, quantitative results, comparisons with published benchmarks, limitations of the current approach, and recommendations for future research directions.

2. Problem Statement

Autonomous vehicle systems depend critically on real-time, accurate perception of the surrounding environment. Image classification — the task of correctly assigning a label to an input image from a fixed set of categories — sits at the foundation of object recognition pipelines used in self-driving technology. Errors in this layer propagate downstream into path planning and control systems, making classification accuracy a safety-critical requirement, not merely a product quality metric.

The specific business problem motivating this project is a make-versus-buy decision facing a self-driving car startup. The company has been offered a proprietary image classification algorithm from a vendor called Detectocorp, whose system achieves 70% classification accuracy on the CIFAR-10 benchmark dataset. The startup's leadership must decide whether to license this external algorithm or invest in building an equivalent or superior capability in-house using the team's existing machine learning expertise.

CIFAR-10 serves as the evaluation domain for this decision. The dataset, introduced by Krizhevsky, Nair, and Hinton in 2009, consists of 60,000 colour images across 10 mutually exclusive object classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. These classes represent categories of objects that an autonomous vehicle system would commonly encounter in real-world driving environments, making performance on CIFAR-10 a meaningful, if simplified, proxy for real-world perception capability.

The classification problem presents several inherent challenges that make 70% accuracy a modest rather than impressive benchmark. CIFAR-10 images are low resolution (32×32 pixels), which limits the availability of discriminative visual features. Several class pairs — notably cat/dog and automobile/truck — share significant visual similarity, creating systematic confusion. Intra-class variation is high, as objects appear across diverse viewpoints, lighting conditions, and backgrounds. These factors collectively make the top-1 accuracy curve on CIFAR-10 a meaningful indicator of a model's capacity to learn robust, generalisable visual representations.

The project therefore tasks the team with designing, training, and evaluating a neural network architecture capable of surpassing 70% accuracy on CIFAR-10. Success demonstrates that the in-house team can deliver a superior product to the Detectocorp offering, informing the strategic build-versus-buy decision. An optional standout objective further challenges the team to implement transfer learning techniques on top of pre-trained networks to push accuracy to the highest level achievable within the project scope.

3. Dataset

The CIFAR-10 dataset (Canadian Institute For Advanced Research, 10 classes) was created by Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton and remains one of the most widely used benchmarks in computer vision research. The dataset is publicly available through the torchvision library and is automatically downloaded during project setup.

3.1 Dataset Composition

CIFAR-10 contains 60,000 colour images divided into a training set of 50,000 images and a test set of 10,000 images. The ten classes are: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The class distribution is perfectly balanced at 6,000 images per class in the training set and 1,000 images per class in the test set. Each image is a 32×32 pixel RGB photograph encoded as three channels of uint8 values in the range [0, 255].

3.2 Data Preprocessing

Two distinct preprocessing pipelines were applied: one for the CIFARNet model and one for the ResNet-50 transfer learning model, reflecting their different architectural requirements.

For CIFARNet, images were normalised using CIFAR-10 channel statistics (mean: [0.4914, 0.4822, 0.4465]; standard deviation: [0.2023, 0.1994, 0.2010]), which centres the pixel distribution around zero and scales it to unit variance. This normalisation stabilises gradient flow and accelerates convergence. The training pipeline additionally applied four data augmentation techniques: random horizontal flipping ($p=0.5$), random cropping with four-pixel padding, colour jitter (brightness, contrast, and saturation perturbation of $\pm 20\%$), and random rotation up to 15 degrees.

For ResNet-50 transfer learning, images were upsampled from 32×32 to 224×224 pixels using bilinear interpolation, matching the spatial resolution on which the ImageNet-pretrained backbone was originally trained. ImageNet channel statistics (mean: [0.485, 0.456, 0.406]; std: [0.229, 0.224, 0.225]) were applied to ensure the pixel distribution matched the pretrained model's expected input range. Augmentation was also applied at 224×224 scale: random horizontal flip, random crop with 28-pixel padding, colour jitter with $\pm 30\%$ perturbations, and random rotation up to 15 degrees.

3.3 Data Loader Configuration

Parameter	CIFARNet Loader	ResNet-50 Loader
Batch size	128	64
Train shuffle	True	True

Parameter	CIFARNet Loader	ResNet-50 Loader
num_workers	2	4
pin_memory	False	True
Train batches	391	782
Test batches	79	157
Input resolution	3 × 32 × 32	3 × 224 × 224

4. Solution Statement

The solution to the CIFAR-10 image classification problem was implemented in two complementary stages. The first stage produced CIFARNet, a custom residual convolutional neural network designed and trained from scratch. The second stage applied transfer learning to a ResNet-50 backbone pre-trained on ImageNet, fine-tuning it for the CIFAR-10 task using a structured two-stage protocol. Both stages used the PyTorch deep learning framework with GPU acceleration.

4.1 Training Framework

All experiments were conducted using PyTorch 2.x with CUDA-enabled GPU acceleration on an NVIDIA Tesla T4 GPU (15.6 GB VRAM) running through Google Colab. The choice of PyTorch was driven by its dynamic computation graph, extensive documentation, and first-class support for both custom model definition and pretrained model libraries via torchvision.

4.2 CIFARNet Training Protocol

CIFARNet was trained for 30 epochs using Stochastic Gradient Descent (SGD) with momentum 0.9 and weight decay 5×10^{-4} . Weight decay provides L2 regularisation, penalising large weight magnitudes and reducing overfitting. The initial learning rate was set to 0.1 — a standard starting point for SGD on CIFAR-scale problems. A Cosine Annealing learning rate scheduler was applied over 30 epochs, smoothly reducing the learning rate from 0.1 to near zero following a cosine curve. This schedule has been empirically shown to help models converge to flatter, more generalisable minima compared to step-decay schedules. Training loss was computed using Negative Log-Likelihood Loss (NLLLoss), which pairs with the log-softmax output activation of CIFARNet and is mathematically equivalent to Cross-Entropy Loss.

4.3 Transfer Learning Protocol

Transfer learning was implemented in two sequential stages designed to prevent catastrophic forgetting of the pretrained ImageNet features. In Stage 1, the entire ResNet-50 backbone was frozen (99.9% of parameters set to `requires_grad=False`), and only the newly added 10-class classification head was trained for five epochs using the Adam optimiser at a learning rate of 1×10^{-3} . This head-warmup stage allowed the classifier to reach a reasonable initialisation before the backbone was unfrozen.

In Stage 2, all 23.5 million parameters were unfrozen and trained for 15 additional epochs using SGD with momentum 0.9. Differential learning rates were applied: the backbone received a learning rate of 1×10^{-4} to make conservative adjustments to the pretrained features, while the classification head retained a learning rate of 1×10^{-3} to allow continued aggressive adaptation. A OneCycleLR scheduler (Leslie Smith's one-cycle policy) was

applied per batch over Stage 2, ramping the learning rate up from a low starting value before annealing it back down, facilitating what has been termed superconvergence in the literature.

4.4 Regularisation Techniques

Both models employed multiple complementary regularisation strategies to prevent overfitting on the 50,000-image training set. These included: data augmentation (four independent transforms applied stochastically at each training step); batch normalisation after every convolutional layer, which acts as an implicit regulariser by reducing internal covariate shift; dropout ($p=0.3$ for CIFARNet, $p=0.4$ for ResNet-50 head) applied immediately before the final linear classification layer; and weight decay (L2 penalty) in the SGD optimiser.

5. Architecture

5.1 CIFARNet — Custom Residual Network

CIFARNet is a compact, purpose-built residual convolutional neural network designed specifically for the CIFAR-10 task. Its design philosophy balances representational capacity against parameter efficiency, targeting a network capable of high accuracy without the computational cost of large-scale architectures.

The architecture consists of four principal components. The stem is a single 3×3 convolution that maps the 3-channel RGB input to 64 feature maps, followed by batch normalisation and ReLU activation. The residual body is structured as three sequential stages. Stage 1 applies two residual blocks at 64 channels. Stage 2 applies two residual blocks at 128 channels, with the first block using a stride of 2 to halve the spatial resolution from 32×32 to 16×16 while doubling the channel count. Stage 3 applies two residual blocks at 256 channels, again with stride-2 downsampling, reducing spatial resolution to 8×8 . The total body therefore contains six residual blocks and twelve convolutional layers.

Each residual block follows the standard pre-activation residual design: two 3×3 convolutions each preceded by batch normalisation and ReLU, with a parallel shortcut connection. When the block changes the number of channels or spatial dimensions, the shortcut uses a 1×1 convolution with matching stride and batch normalisation to project the identity to the correct shape before the element-wise addition. This skip connection mechanism allows gradients to flow directly through the network depth without attenuation, addressing the vanishing gradient problem that limits the depth of plain networks.

The classifier head uses Global Average Pooling (GAP) to collapse the 256-channel 8×8 feature maps to a 256-dimensional vector, followed by Dropout ($p=0.3$) and a single fully-connected linear layer projecting to 10 class logits. Log-softmax is applied to produce log-probability outputs compatible with NLLLoss.

Component	Operation	Output Shape
Stem	Conv2d(3→64, k=3, pad=1) + BN + ReLU	$64 \times 32 \times 32$
Stage 1 Blk 1	ResidualBlock(64→64, stride=1)	$64 \times 32 \times 32$
Stage 1 Blk 2	ResidualBlock(64→64, stride=1)	$64 \times 32 \times 32$
Stage 2 Blk 1	ResidualBlock(64→128, stride=2) + shortcut	$128 \times 16 \times 16$
Stage 2 Blk 2	ResidualBlock(128→128, stride=1)	$128 \times 16 \times 16$

Component	Operation	Output Shape
Stage 3 Blk 1	ResidualBlock(128→256, stride=2) + shortcut	256 × 8 × 8
Stage 3 Blk 2	ResidualBlock(256→256, stride=1)	256 × 8 × 8
Head	GlobalAvgPool → Dropout(0.3) → Linear(256→10)	10

Total trainable parameters: 2,777,674 (~2.8 million). The final test accuracy achieved was 93.43%.

5.2 ResNet-50 Transfer Learning Architecture

ResNet-50 is a 50-layer deep residual network originally designed by He et al. (2016) for the ImageNet Large Scale Visual Recognition Challenge. It uses a bottleneck residual block design with 1×1, 3×3, and 1×1 convolutional layers, organised into four stages of increasingly deep feature maps (64, 128, 256, and 512 channels), followed by Global Average Pooling and a 2048→1000 linear classifier.

For CIFAR-10, the pretrained ImageNet classifier head (Linear(2048, 1000)) was replaced with a new head consisting of Dropout (p=0.4) followed by a Linear(2048, 10) projection layer. The backbone weights were initialised from the IMAGENET1K_V2 checkpoint, which achieves 80.85% top-1 accuracy on ImageNet — a dramatically richer initialisation than random weights. The total parameter count of the adapted model is 23,528,522, of which only 20,490 (0.1%) were trainable in Stage 1.

Because CIFAR-10 images are 32×32 but ResNet-50 expects 224×224 inputs, images were upsampled using bilinear interpolation before entering the network. This ensures the pretrained spatial feature maps — calibrated for 224×224 input — function as intended. The final test accuracy of the fine-tuned model was 94.62%.

6. Comparison: CIFARNet, ResNet-50, and Published Benchmarks

The following table places the two models developed in this project in context against five published CIFAR-10 benchmark results cited in the project brief, as well as the Detectocorp commercial baseline. Results are sorted by test accuracy.

Model	Authors / Source	Year	Test Accuracy	Notes
Deep Belief Networks	Krizhevsky	2010	78.9%	Early deep generative approach; no GPU-scale conv nets
Detectocorp (baseline)	Commercial vendor	N/A	70.0%	Proprietary; below academic baselines from 2010
Maxout Networks	Goodfellow et al.	2013	90.6%	Introduced maxout activation for competitive CIFAR accuracy
CIFARNet (this project)	In-house (custom CNN)	2026	93.43%	2.8M params; trained from scratch; 30 epochs on CIFAR-10
ResNet-50 TL (this project)	In-house (fine-tuned)	2026	94.62%	23.5M params; ImageNet pretrained; 20 epochs fine-tuning
Wide Residual Networks	Zagoruyko et al.	2016	96.0%	Wider, shallower residual nets; strong CIFAR baseline
Rethinking RNNs	Nguyen et al.	2020	98.5%	Hybrid attention-recurrent architecture; near-SOTA
GPipe	Huang et al.	2018	99.0%	Pipeline-parallel training at massive scale; SOTA at time

6.1 Analysis

Both models developed in this project surpass the Detectocorp commercial baseline by a substantial margin and comfortably exceed the Maxout Networks result from 2013. CIFARNet, with only 2.8 million parameters trained from scratch, achieves 93.43% — a result competitive with many published architectures and within three percentage points of the Wide Residual Network result using far fewer parameters and a simpler training setup.

The ResNet-50 fine-tuned model achieves 94.62%, which further closes the gap to the Wide Residual Network benchmark (96.0%) while using only 20 epochs of task-specific training. This demonstrates the substantial leverage that transfer learning provides: rather than training 23.5 million parameters from random initialisation, the vast majority of learning had already been completed on ImageNet, requiring only targeted adaptation to CIFAR-10.

The gap between the in-house models and the SOTA results (GPipe at 99.0%, Rethinking RNNs at 98.5%) reflects the scale of infrastructure and data required at the frontier. GPipe was trained across hundreds of accelerators using pipeline parallelism and enormous batch sizes, while advanced architectures like the one in Nguyen et al. (2020) integrate attention mechanisms and multi-stage training pipelines well beyond the scope of a single-project implementation. Reaching these accuracy levels would require either training at cloud-scale or using more advanced pretrained foundation models such as Vision Transformers (ViT) or EfficientNet-B7.

Importantly, the in-house models already exceed the Detectocorp baseline by 23.43 percentage points (CIFARNet) and 24.62 percentage points (ResNet-50), representing a dramatic relative improvement. From a business perspective, this margin is more than sufficient to justify the build decision on both technical and commercial grounds.

6.2 Per-Class Performance

Class	CIFARNet Accuracy	ResNet-50 Accuracy	Harder / Easier
Airplane	94.80%	94.60%	Easier
Automobile	96.90%	97.70%	Easier
Bird	90.50%	92.90%	Moderate
Cat	86.30%	88.20%	Hardest (visual similarity to dog)
Deer	94.00%	93.90%	Easier
Dog	88.80%	91.90%	Hard (visual similarity to cat)
Frog	96.20%	95.60%	Easier
Horse	95.10%	96.50%	Easier
Ship	96.10%	97.40%	Easier
Truck	95.30%	96.80%	Easier

The cat and dog classes consistently represent the greatest classification challenge across both models, reflecting their shared morphological features at 32×32 resolution. This pattern

is well-documented in the CIFAR-10 literature and represents a fundamental limit of the low-resolution input rather than an architectural deficiency. Vehicle classes (automobile, ship, truck) and distinctive animal silhouettes (frog, horse) achieve the highest per-class accuracy.

7. Conclusion

This project successfully demonstrated that an in-house deep learning team can design, implement, and train image classification models that substantially outperform commercial vendor offerings. Two models were developed using the PyTorch framework: CIFARNet, a custom 2.8-million-parameter residual CNN trained from scratch, achieved 93.43% test accuracy on CIFAR-10 after 30 epochs; and a fine-tuned ResNet-50, leveraging ImageNet pretrained weights through a structured two-stage transfer learning protocol, achieved 94.62% accuracy after 20 epochs.

Both models exceeded the 70% Detectocorp commercial baseline by more than 23 percentage points, surpassed the 45% minimum project requirement by a wide margin, and produced competitive results relative to published academic benchmarks from the past decade. The per-class accuracy analysis confirmed consistently strong performance across all ten CIFAR-10 categories, with cat and dog classification representing the most challenging cases due to their visual similarity at low resolution — a known, documented limitation of the dataset rather than a model deficiency.

The training methodology combined multiple complementary techniques to achieve these results: data augmentation (horizontal flip, random crop, colour jitter, random rotation) to improve generalisation; batch normalisation throughout all convolutional stages for training stability; residual skip connections to enable effective gradient propagation through depth; cosine annealing learning rate scheduling for CIFARNet; and differential learning rates with OneCycleLR scheduling for ResNet-50 Stage 2 fine-tuning. The combined use of these techniques represents current best practice in CNN training for medium-scale image classification.

Transfer learning emerged as a particularly powerful technique in this project context. By leveraging the rich visual representations learned from 1.2 million ImageNet images, the ResNet-50 model reached 83.25% accuracy within five epochs of head-only training — before any backbone fine-tuning had occurred. This rapid initial convergence illustrates that ImageNet features are broadly transferable to CIFAR-10, consistent with the observation that both datasets involve natural photographic images of real-world objects.

The overall conclusion is clear: the in-house approach is both technically superior and strategically preferable. The models produced in this project outperform the Detectocorp offering significantly, provide full transparency and control over the architecture and training pipeline, and establish a foundation for continued improvement as proprietary operational data becomes available from the vehicle fleet.

8. Value Statement

The results produced in this project deliver clear and multi-dimensional value to a self-driving car startup facing the build-versus-buy decision. This section articulates the key dimensions of that value across technical, commercial, operational, and strategic axes.

8.1 Technical Superiority

The in-house CIFARNet model achieves 93.43% accuracy compared to Detectocorp's 70%, a relative accuracy improvement of approximately 33%. The ResNet-50 transfer learning model further extends this to 94.62%. In safety-critical autonomous vehicle applications, the difference between 70% and 93–94% accuracy is not merely a metric improvement — it translates to a reduction in misclassification events of approximately 78%, directly impacting the safety of both passengers and road users. No commercial procurement can offer the combination of tailored performance and architectural transparency that an in-house solution provides.

8.2 Cost and Ownership

Purchasing the Detectocorp algorithm introduces recurring licensing costs, contractual dependencies, and limited ability to audit or modify the underlying system. The in-house approach, once developed, has zero marginal cost per inference and is fully owned intellectual property. The capital expenditure for GPU compute time on cloud platforms (Google Colab, AWS, or Google Cloud) required to train both models is minimal compared to enterprise software licensing fees, while producing substantially superior results. Ownership of the model weights and architecture also protects against vendor lock-in and enables the company to pivot its classification strategy as requirements evolve.

8.3 Adaptability and Continuous Improvement

Perhaps the most important value proposition of the in-house approach is adaptability. The transfer learning framework demonstrated in the ResNet-50 standout section establishes an efficient pipeline for fine-tuning on new data. As the vehicle fleet accumulates real-world road imagery — which will differ from CIFAR-10 in resolution, camera perspective, and object distribution — the in-house model can be continuously retrained on this proprietary data to improve accuracy in the actual deployment domain. No external vendor can replicate this feedback loop, as they lack access to the company's operational data.

8.4 Intellectual Property and Competitive Moat

A high-performing, proprietary perception model is a genuine competitive moat in the autonomous vehicle industry. Demonstrating that the in-house team can develop state-of-the-art classification technology signals technical capability to investors, partners, and

potential acquirers. It also preserves the option to commercialise the technology as a standalone product or service in the future — an option that is entirely foreclosed if the company depends on a third-party vendor's black-box algorithm.

9. Limitations

Despite the strong results achieved in this project, several important limitations must be acknowledged when interpreting the findings and considering their applicability to real-world autonomous vehicle deployment.

9.1 Dataset Resolution and Realism

CIFAR-10 images are 32×32 pixels — a resolution far below that of cameras used in actual self-driving systems, which typically produce images at 1280×720 pixels or higher. While CIFAR-10 provides a useful research benchmark, the classification task it poses does not fully represent the challenges of real automotive perception, which involves detecting and classifying objects at varying scales, distances, and occlusion levels within high-resolution frames. Models trained on CIFAR-10 cannot be deployed directly in an autonomous vehicle without substantial retraining on domain-appropriate data.

9.2 Closed-Set Classification

Both models perform closed-set classification: they assign each input to one of exactly ten predefined categories. Real-world autonomous perception requires open-set recognition — the ability to detect and flag objects that belong to categories not seen during training (for example, construction equipment, emergency vehicles, or unusual road obstacles). Neither CIFARNet nor the fine-tuned ResNet-50 has this capability, and deploying a closed-set classifier in an open-world environment introduces the risk of confidently misclassifying unknown objects.

9.3 Potential Overfitting in Transfer Learning Stage 1

The rapid jump to 82.05% accuracy in the very first epoch of ResNet-50 Stage 1 (with the backbone frozen) is impressive, but may reflect a degree of feature overfitting to the CIFAR-10 training distribution given the limited expressive capacity of the head alone. The subsequent plateau at approximately 83–84% before Stage 2 fine-tuning suggests that the frozen backbone, while highly informative, cannot fully adapt its spatial feature detectors to the 224×224 upsampled CIFAR-10 images without parameter updates.

9.4 Hardware and Scale Constraints

Both models were trained on a single NVIDIA Tesla T4 GPU with 15.6 GB VRAM. The most accurate results on CIFAR-10 in the literature (GPipe at 99.0%) were achieved using pipeline-parallel training across hundreds of accelerators with extremely large batch sizes. The performance gap between the in-house models (~93–94%) and these frontier results reflects a hardware and infrastructure constraint rather than an algorithmic one. Closing this gap would require access to significantly greater compute resources.

9.5 Single Random Seed

Results reported in this project reflect single training runs without cross-validation or repeated experiment averaging. Training deep neural networks with stochastic optimisation introduces variance from random weight initialisation, data shuffling, and data augmentation sampling. The reported accuracy figures may therefore differ by one to two percentage points across independent runs, and the true expected accuracy should be interpreted as a distribution rather than a point estimate.

10. Directions for Future Research

The results achieved in this project open several productive directions for future investigation, both in the context of improving CIFAR-10 benchmark performance and in extending the work toward deployment in real autonomous vehicle perception systems.

10.1 Advanced Architecture Exploration

The current models, while competitive, have not reached the state of the art on CIFAR-10 (99.0% with GPipe). Future work could explore more powerful backbone architectures, including EfficientNet (Tan and Le, 2019), which achieves better accuracy-efficiency trade-offs through compound scaling of width, depth, and resolution; Vision Transformers (ViT, Dosovitskiy et al., 2020), which have demonstrated that transformer-based architectures can match or surpass CNNs on image classification; and DeiT (Data-efficient Image Transformers, Touvron et al., 2021), which adapts the ViT framework for settings with limited training data through knowledge distillation — particularly relevant for fine-tuning on smaller proprietary datasets.

10.2 Knowledge Distillation

The ResNet-50 fine-tuned model achieves 94.62% accuracy but requires 23.5 million parameters — a significant inference cost for a real-time embedded system. Knowledge distillation (Hinton et al., 2015) offers a method to transfer the predictive performance of this large teacher model into a much smaller student model (such as CIFARNet with its 2.8 million parameters) by training the student on the soft probability outputs of the teacher rather than hard class labels. This approach can recover a substantial fraction of the teacher's accuracy in a fraction of the parameter budget, which is critical for real-time deployment on vehicle-embedded hardware.

10.3 Domain Adaptation to Real Road Imagery

The most impactful future direction for the self-driving application is fine-tuning on proprietary road imagery collected from the vehicle fleet. As the company accumulates operational data, the ResNet-50 backbone — already primed on both ImageNet and CIFAR-10 — can be further fine-tuned using domain adaptation techniques. Methods such as self-supervised pre-training on unlabelled road imagery (using contrastive learning frameworks such as SimCLR or MoCo) followed by supervised fine-tuning on manually labelled frames could substantially improve accuracy in the deployment domain.

10.4 Object Detection and Segmentation

CIFAR-10 is a whole-image classification task, but real autonomous vehicle perception requires localised object detection (bounding boxes) and semantic segmentation (pixel-level

labelling). Future work should extend the classification models developed here into detection frameworks such as YOLO (Redmon et al., 2016) or Faster R-CNN (Ren et al., 2015), or segmentation models such as DeepLab or Mask R-CNN. The ResNet-50 backbone fine-tuned in this project can serve directly as the feature extractor backbone for these detection and segmentation heads, making the current work a natural stepping stone toward full scene understanding.

10.5 Uncertainty Quantification

Deployment in safety-critical autonomous systems requires not just accurate predictions, but well-calibrated confidence estimates. Current models produce point predictions without uncertainty bounds. Future research should investigate Bayesian deep learning approaches (Monte Carlo Dropout, Deep Ensembles) or post-hoc calibration methods (temperature scaling, Platt scaling) to equip the classifier with reliable confidence scores. This would allow the autonomous system to flag low-confidence predictions for human review or fallback to conservative driving behaviours, substantially improving the safety profile of the deployed system.

11. References

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— *End of Report* —